Does the Credit Market Price Transition Risk at the Wrong Maturity?

Carbon drainage, distance to default, and the term structure of credit spreads — Scoping note and literature review

Josh SOUSSAN - Edouard SOUIED - Nicolas SALAPETE - Raphaël HAZAN
CentraleSupelec - ESSEC

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Abstract

Climate transition risk differs from most credit risk factors in its *temporal signature*: it acts as a slow and increasing erosion of cash flows rather than as an immediate shock. We derive a testable structural prediction from this observation. In a Merton (1974) framework, adding a carbon drain whose intensity rises over time distorts the term structure of credit spreads in a strongly asymmetric way: holding leverage and asset volatility constant, a carbon-intensive firm sees its 5–10 year slope widen by almost 80%, against only 21% for its five-year spread level. Yet the empirical literature and market practice test almost exclusively the five-year point, which is precisely the maturity at which the signal is most diluted. We propose to test the slope prediction directly, using the five-year level as a placebo test, and to identify the effect through surprises in the price of European carbon allowances (EUAs).

1 Research Question

1.1 Statement

Does the credit market price transition risk at the wrong maturity?

Stated as a testable hypothesis: if transition risk is a gradual drain on cash flows rather than a risk of imminent default, then it should appear primarily in the *slope* of the CDS curve rather than in its five-year *level*. Because the literature and practitioners overwhelmingly observe the five-year point — the benchmark contract and the most liquid point on the curve — they measure the signal where it is structurally weakest.

1.2 Sub-questions

Q1. **Theory.** What distortion of the term structure of spreads follows from introducing a rising carbon drain into a Merton-type firm value model? What is the relative magnitude of the effect on the slope versus the level?

Q2. **Empirics.** Matched on financial profile (leverage, size, rating, sector), is the CDS curve of carbon-intensive issuers significantly steeper? Is the effect, as predicted, substantially weaker on the five-year level than on the slope?

Q3. **Identification.** When the EUA price is subject to a surprise, does the curve of the most exposed issuers steepen more than that of their matched peers?

2 Theoretical Framework and Prediction

2.1 Merton with a carbon drain

In the structural model of Merton (1974), firm asset value follows

$$dV_t = \mu V_t\,dt + \sigma V_t\,dW_t, \tag{1}$$

and default occurs if $V_T < D$, where D is the face value of debt. A firm emitting E tonnes of CO_2 per year, facing a carbon price $P_c(t)$ and able to pass through a fraction ρ of that cost to its customers, bears a net annual drain scaled by asset value:

$$\delta(t) = \frac{P_c(t)\,E\,(1-\rho)}{V}. \tag{2}$$

This drain is subtracted from the drift, $dV_t = (\mu - \delta(t))V_t\,dt + \sigma V_t\,dW_t$, yielding a modified distance to default

$$DD(T) = \frac{\ln(V_0/D) + (\mu - \bar{\delta}(T) - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad \bar{\delta}(T) = \frac{1}{T}\int_0^T \delta(s)\,ds, \tag{3}$$

and a credit spread, under the risk-neutral measure and with recovery rate R:

$$s(T) = -\frac{1}{T}\ln\Big(1 - (1-R)\,\mathcal{N}\big(-DD^{\mathbb{Q}}(T)\big)\Big). \tag{4}$$

2.2 Where the asymmetry comes from

The mechanism is entirely contained in the $\bar{\delta}(T)$ term of equation (3). All credible climate policy trajectories imply $P_c'(t) > 0$: the carbon price is low today and rising. Consequently $\bar{\delta}(T)$ is increasing in T, and the cumulative drain is negligible at short horizons but comparable to the solvency cushion at long horizons. Carbon therefore distorts the curve by construction, rather than shifting it.

2.3 Illustrative calibration

We simulate two firms that are strictly identical in financial terms ($V_0/D = 2.5$, $\sigma = 26\%$, $r = 3\%$, $R = 40\%$), differing only in δ_0, the initial net carbon drain. The carbon price starts at €80/t and grows at 7.5% per year, reaching roughly €250/t by 2040; the pass-through rate is set at $\rho = 0.4$.

Table 1: Simulated spreads by maturity (basis points)

Maturity	1y	2y	3y	5y	7y	10y
Low carbon intensity	1.4	20.4	45.2	75.8	87.6	90.8
High carbon intensity	1.6	23.8	53.4	92.0	109.4	118.9
Difference	0.2	3.4	8.2	16.2	21.8	28.0

The central prediction can be read directly off Figure 1 and Table 1: **the carbon signal is roughly twice as strong in the slope as in the five-year level.** This asymmetry is the quantity we propose to test.

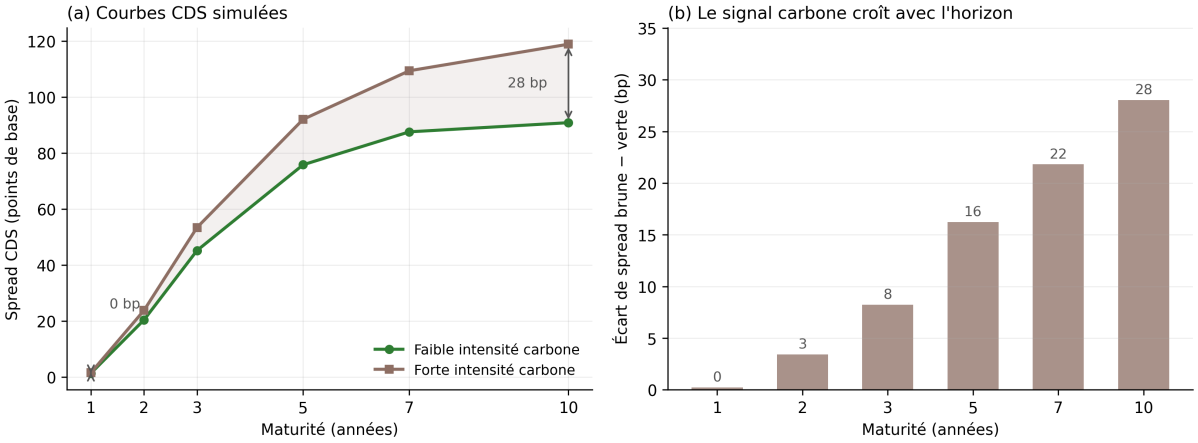


Figure 1: **Effect of a rising carbon drain on the term structure of CDS spreads.** Panel (a): simulated curves for two firms identical in leverage, volatility and size, differing only in carbon intensity. Panel (b): spread differential between the two firms by maturity. The gap widens from under 1 bp at one year to 28 bp at ten years. The 5–10 year slope rises from 15 to 27 bp (+79%), while the five-year level rises by only 21%.

3 Literature Review

We organise the literature into four blocks: structural credit models and their extension to carbon, single-horizon empirical tests, work that explicitly exploits the CDS term structure, and finally the conflicting results that delimit our contribution.

3.1 Structural models and the carbon extension

The starting point is Merton (1974), who establishes that corporate debt can be valued as an option on firm assets, and in doing so delivers an endogenous term structure of credit spreads. It is this property, often overlooked in climate applications, that we exploit.

Le Guenedal and Tankov (2025) develop the most complete extension of this framework to transition risk. They build a structural model for pricing a defaultable bond issued by a firm subject to a carbon tax whose trajectory depends on a transition scenario that is initially unknown and progressively revealed through observation of the carbon price. Bond price, credit spread, and optimal default or restructuring thresholds are then expressed as functions of the firm's revenue level and the carbon price. Their central result is that scenario uncertainty makes carbon tax adjustments more likely to trigger default than when the scenario is known. This work is closest to ours theoretically, but it remains a calibrated valuation exercise: it derives no signed prediction about the slope, and does not confront one with market data. Our contribution is precisely to turn this family of models into a discriminating empirical prediction.

3.2 Single-horizon empirical tests

Capasso et al. (2020) provide the benchmark empirical result linking carbon to credit risk. They show that Merton's distance to default is negatively associated with the level and intensity of a firm's emissions, and that this relationship strengthens following shocks revealing policymakers' intent to tighten climate policy, notably the Paris Agreement. Their measure is, however, a scalar: distance to default is estimated at a single horizon, conventionally one year. By construction, such a measure cannot detect the asymmetry our model predicts, since it collapses

information from the entire curve into a single point. We consider it important to state this limitation explicitly, because it is shared by much of the subsequent literature.

Bolton and Kacperczyk (2021) provide the benchmark result on the equity side: returns of high-emission firms are significantly higher, which the authors interpret as a carbon risk premium demanded by investors. This motivates the existence of a priced carbon factor, but says nothing about its distribution along the term structure, nor about its transmission to the credit market.

3.3 Work exploiting the term structure

Three recent papers address the temporal dimension directly, which makes our question both relevant and contested.

Kölbel et al. (2024) use BERT to quantify regulatory climate risk disclosures and measure their effect on the term structure of the CDS market. They show that a disclosure can raise or lower spreads depending on whether it reveals new risk or reduces uncertainty, and that distinguishing transition from physical risk, transition risk disclosure raises spreads after the Paris Agreement while physical risk disclosure lowers them. Their explanatory variable is textual, therefore forward-looking but noisy, and their object is the effect of a disclosure rather than physical exposure to carbon. The term structure is for them a dimension of analysis, not a prediction derived from a model.

Ugolini et al. (2024) construct a transition risk factor from the vulnerability of firm value to the low-carbon transition, and document that this factor shifts the term structure of CDS spreads for the most vulnerable firms, with no significant effect on the others. They further show that different transition policies have asymmetric and economically significant effects on the credit risk of vulnerable firms, and negligible effects on the rest. This is the work whose conclusions come closest to ours, but their vulnerability measure is constructed from asset prices themselves, which raises a circularity concern: the factor is extracted from markets, then tested on markets. Our approach instead starts from verified emissions and an explicit structural mechanism.

Blasberg et al. (2024) offer the result most directly at odds with our prediction, and this is what makes the project interesting. Analysing the relationship between CDS term structures and carbon risk, they conclude that an increase in carbon risk leads lenders to anticipate near-term transition costs, translating into a more pronounced rise in spreads at shorter tenors than at longer ones, signalling market anticipation of immediate financial consequences. They further document that an explicit carbon pricing regime triples the additional cost of credit protection, and that the effect is stronger for European than for North American firms.

3.4 The empirical disagreement, and what it opens

The state of the literature thus presents a clear tension. A structural model with a rising carbon price mechanically implies long-maturity dominance, a prediction we quantify in Section 2. Ugolini et al. (2024) document a shift in the term structure concentrated among vulnerable firms, without settling the maturity question as sharply. Blasberg et al. (2024) find, on the contrary, short-maturity dominance.

Three readings of this disagreement are possible, and all are testable. First, the market may be anticipating an abrupt regulatory shock rather than a gradual rise in the carbon price, in which case a jump model is more appropriate than a drift model, and the maturity signature reverses. Second, the discrepancy may stem from the carbon risk measure used: a measure extracted from asset prices inherits the short-run dynamics of markets, whereas verified emissions are a slow-moving variable. Third, the long end of the CDS curve is illiquid outside financials, and a substantial share of ten-year quotes is interpolated by data providers rather than actually

traded, which mechanically attenuates any signal measured at that maturity.

This last reading is the one we find most promising, because it is both an explanation of the disagreement and a result in itself: if transition risk sits precisely at the least liquid maturity of the curve, there is a structural reason for it to remain imperfectly priced.

3.5 Positioning

Our contribution differs on three counts. First, we derive a *signed and quantified prediction, prior to observing the data*, about the ratio of the slope effect to the level effect, whereas the empirical literature proceeds without a prior structural prediction. Second, we build a placebo test into the design: the effect should be weak on the five-year level, and a uniform effect along the curve would constitute a refutation rather than a confirmation. Third, we treat long-end liquidity as a variable of interest rather than a nuisance, which offers a possible reconciliation of the literature's divergent results.

4 Proposed Empirical Strategy

4.1 Sample construction

Within-sector matching (NACE 3- or 4-digit) of issuers close on leverage, size, rating and issuance currency, and distant on Scope 1+2 carbon intensity. Natural candidates: cement against lightweight materials, primary steelmaking against electric arc furnaces, coal-based against renewables-based utilities.

4.2 Specifications

In the cross-section, with $\text{Slope}_i = s_i^{10y} - s_i^{5y}$:

$$\text{Slope}_i = \alpha + \beta\,\text{CI}_i + \gamma' X_i + \eta_{\text{sector}} + \varepsilon_i, \qquad H_1 : \beta > 0, \tag{5}$$

where CI_i is carbon intensity and X_i the financial controls. The placebo test consists in re-estimating the same specification with s_i^{5y} as the dependent variable: the model predicts a markedly weaker coefficient.

In changes, for identification:

$$\Delta\text{Slope}_{i,t} = \alpha + \beta\,(\text{CI}_i \times \Delta\text{EUA}_t) + \gamma\,\Delta\text{EUA}_t + \theta' Z_{i,t} + \varepsilon_{i,t}. \tag{6}$$

The coefficient of interest is the interaction term. We favour EUA price *surprises* — residuals from a forecasting model, or narrow-window changes around European legislative dates — in order to rule out the correlation between carbon price, gas price, and the industrial cycle.

4.3 Data

Multi-maturity CDS curves via Markit, Bloomberg (`CDSD`) or Refinitiv; fallback on bond spreads reconstructed by maturity. Installation-level verified emissions via the EU Transaction Log, supplemented by CDP. Daily EUA prices from open sources. Fundamentals via Compustat Global or Refinitiv.

4.4 Identified limitations

1. **Long-end liquidity.** A share of ten-year quotes is provider interpolation. Filtering on the number of contributors is essential.

2. **Carbon/sector confounding.** Because carbon intensity is strongly correlated with sector, sector fixed effects and matching are required; without them, one merely measures a sectoral regularity with no informational content.

3. **Unobserved pass-through.** ρ enters equation (2) without being observable. We calibrate it by sector from the existing literature and report the sensitivity of the prediction over $\rho \in [0, 0.8]$.

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